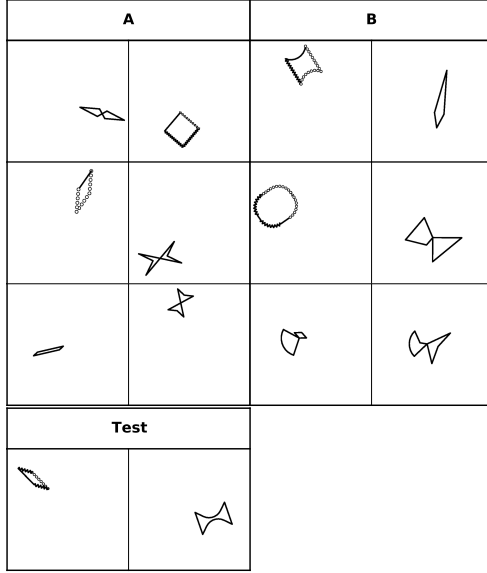
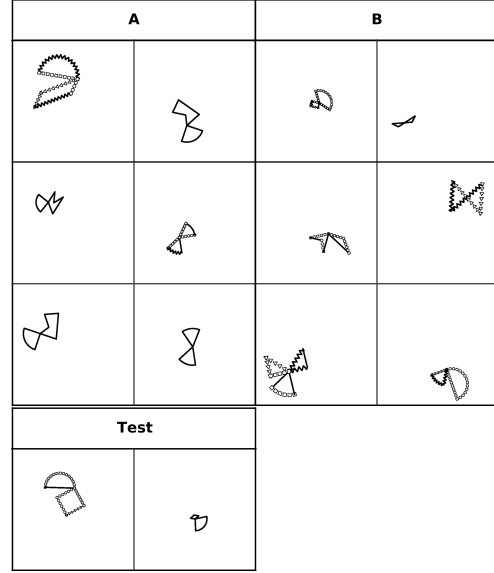
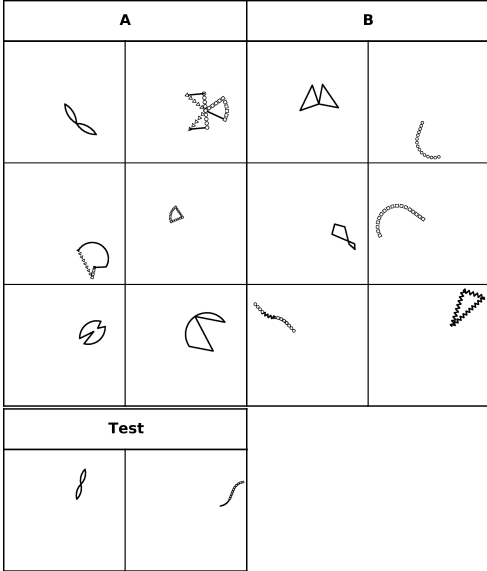




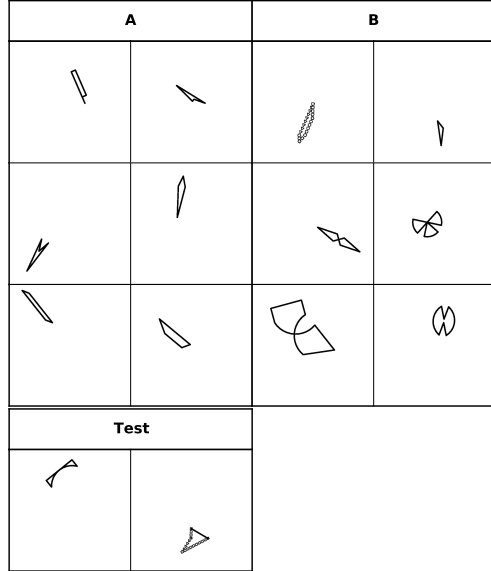
(a) self_transposed and exist_quadrangle



(b) balanced_two and exist_sector



(c) have_curve and closed_shape



(d) have_four_straight_lines and thin_shape

Figure 10: More examples of abstract shape problems, where a combination of two attributes shared by the positive images in set $\mathcal{A}$ is considered as the underlying concept. In particular, (a) the concept is a combination of self_transposed and exist_quadrangle ('self-transposed' means there is a central point, symmetrically around which every edge point of the shape could be mapped to another edge point), (b) the concept is a combination of balanced_two and exist_sector ('balanced' means the two parts in a shape have the similar area), (c) the concept is a combination of have_curve and closed_shape, (d) the concept is a combination of have_four_straight_lines and thin_shape. In each problem, set $\mathcal{A}$ contains six images that satisfy the concept and set $\mathcal{B}$ contains six images that violate the concept. We also show two test images (left: positive, right: negative) in the binary classification problems. We do not distinguish concepts by the shape size and orientation, absolute position, or relative distance of two shapes. We can see for abstract shape problems, it is also challenging for humans without clearly understanding the meanings of abstract attributes.